ALICE L WALTON SCHOOL OF MEDICINE

Chronic Myeloid Leukemia

Last updated November 19, 2025 18 min read ScholarRx

Adapted by: Chelsey Deel, Trager Hintze, Diptiman Bose
Editors: Jordan Finn, Russell Johnson

Adapted from: Chronic Myeloid Leukemia

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of chronic myeloid leukemia (CML), delineating the chronic phase and the blast phase.
2. Explain the pathophysiology of CML.
3. Interpret clinical findings to properly diagnose CML.
4. Outline the management plan of CML.

CASE CONNECTION

“Let me get this straight,” LL says to you. “Your specialist colleague told me that a part of one of my chromosomes went over to join another part of another chromosome, and together, they created a monster. Did I hear her right?” LL is a 52-year-old male who is seeing you in a follow-up to discuss his bone marrow results while your colleague is on vacation. One week ago, LL presented to your partner with fatigue and left upper quadrant pain. She diagnosed LL with chronic myeloid leukemia (CML).

What clues on the complete blood count (CBC) would you expect to see when you review LL’s record? How will you answer LL’s question? Consider your answers as you read, and we’ll revisit LL at the end of the brick.

GO TO CONCLUSION ↓

What Is Chronic Myeloid Leukemia?

Chronic myeloid leukemia (CML) is one among a group of hematopoietic neoplasms called **myeloproliferative neoplasms.** These are a cluster of chronic, slowly progressive diseases in which the bone marrow produces large numbers of neoplastic, mature myeloid cells, which end up spilling out into the blood. In several, one cell line dominates and is present in large numbers in the blood. In CML, it is the **myeloid cell line** (neutrophils, basophils, eosinophils) that undergoes excess proliferation. So, the bone marrow and blood are full of myelocytes and their precursors.

The bottom line is that CML is a chronic myeloproliferative disorder with a prominent neutrophil component. The percentage of immature neoplastic white blood cells am s,” indicates CML progression in the blood. As the disease progresses, the blast count in peripheral blood grows, resulting in an acute medical condition termed “blast crisis” at the end stage.

CML accounts for 15% to 20% of adult leukemias. It usually affects **adults age 50 to 60 years** but can also occur in adolescents and less often in children. Similar to the other myeloproliferative neoplasms, approximately 1 to 2 cases per 100,000 people occur annually.

Myeloproliferative neoplasms are all characterized by overproduction of myeloid cells. In CML, what type of myeloid cell is proliferating the most?

How Do Patients With Chronic Myeloid Leukemia Present Clinically (LO #1)?

Chronic myeloid leukemia can present in a slowly progressive chronic phase, or in a more acute illness termed blast crisis. Most patients present in the chronic phase (85%–90%) and t without treatment, most will develop blast crisis.

Chronic Phase

The chronic phase of CML usually progresses slowly and is relatively stable. Untreated, this phase usually lasts approximately 3 to 4 years; medical treatment can extend it. Patients are often asymptomatic, with the disease being discovered on a routine complete blood count (CBC) done for another reason. When symptoms are present, they may be vague. They could include fatigue or **splenomegaly** symptoms, such as a fullness or dragging feeling in the left upper quadrant, abdominal pain, or early satiety.

Blast Crisis

This phase is basically the same thing as an acute leukemia. As with acute leukemia, the patient may have night sweats, weight loss, fever, and bone pain. Bone marrow failure may give rise to severe weakness and fatigue (resulting from anemia) as well as bleeding (because of low platelets). Without treatment, survival is generally on the order of weeks to a few months.

CLINICAL CORRELATION

In the literature prior to 2022, there is mention of an intermediate “accelerated phase” defined as a state between chronic phase and blast phase. However, the accelerated phase is not included in the fifth edition of the *World Health Organization (WHO) Classification of Haematolymphoid Tumors.*

What are the phases of symptoms that a patient with CML may progress through?

What Is the Pathophysiology of Chronic Myeloid Leukemia (LO #2)?

The only known risk factor for CML is ionizing radiation, and there is no recognized familial inheritance. So, most cases arise from spontaneous gene mutations.

Genetics

All CML cases have a gene mutation termed the *BCR::ABL* fusion gene. Here, the *ABL* oncogene moves from the long arm of chromosome 9 into a new position next to the breakpoint cluster region (*BCR*) gene on the long arm of chromosome 22. The fusion of the *BCR* and *ABL* genes is called the **t(9;22) translocation** and c 1).

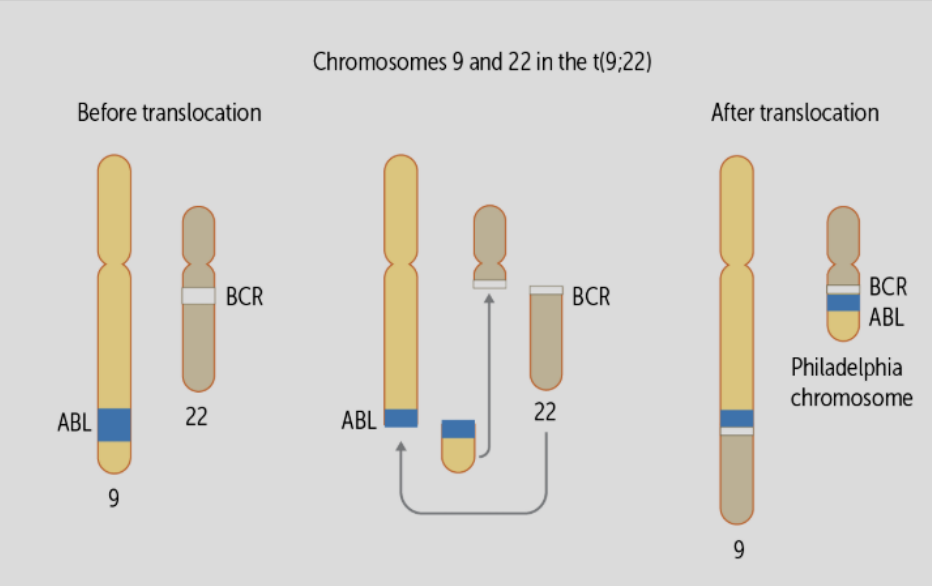

Figure 1

This joining of unrelated genes causes abnormal cell function, such as increased cell proliferation and defective DNA repair. The main mechanism is an overactive tyrosine kinase enzyme. The abnormal fusion gene is known as the Philadelphia chromosome, a characteristic of CML which can be tested for in genetic analysis.

CLINICAL CORRELATION

The Philadelphia ch is also less commonly seen in acute lymphoblastic leukemia and acute myeloid leukemia. So, the presence of the chromosome is not in itself diagnostic of CML.

Of note, the *BCR::ABL1* translocation in ALL is created at a different breakpoint in the *BCR* gene, causing a different sized fusion protein.

CML is characterized by what chromosomal abnormality?

Pathology

Let’s look at blood and bone marrow of the patients in the various stages of CML. The chronic phase is most common (hence the term chronic myeloid leukemia), so we will spend most of our time there.

Chronic Phase. The peripheral blood in the chronic phase shows a slowly increasing **leukocytosis** (high WBC count). The mean count is approximately 100,000 cells/mm^3 (normal range, 4500–11,000 cells/mm^3), but the count can increase much more, up to 1,000,000/mm^3. The leukocytosis is mostly neutrophils but includes a **left shift,** which means that in addition to mature, segmented neutrophils, there are also immature neutrophil precursors, such as myelocytes, that typically are not present in the blood. Notably, the blast count is low (it’s usually somewhere around 2%–3% but not more than 10%). This contrasts CML with acute myeloid leukemia, where blasts are much more elevated (>20%). These neutrophils may appear normal, but they contain the genetic abnormalities discussed above.

In most patients with CML, all myeloid lines increase; therefore, eosinophils and basophils also rise. The **basophilia** is notable, because it’s rare to see a basophil on the CBC (just a little pearl to stick in the back of your head!).

Blood Smear. Let’s take a look at a blood smear in the chronic phase of CML and identify the different cells (Figure 2).

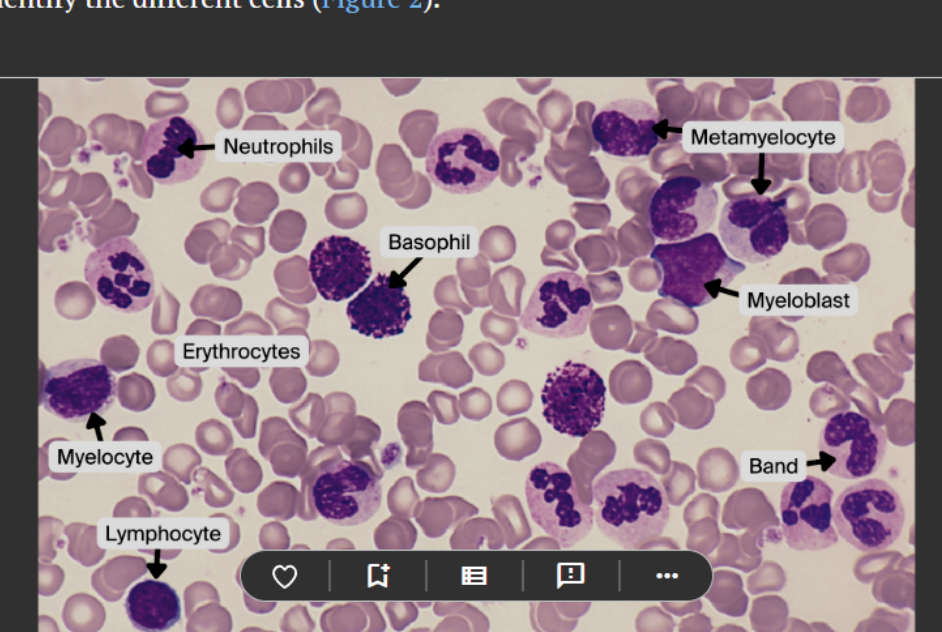

Figure 2: Chronic Myeloid Leukemia Blood Smear

In Figure 2, we see lots of mature neutrophils and metamyelocytes. And check out all the basophils! Notice the very low number of blasts (compare that with acute leukemia, where almost every neoplastic cell is a blast).

Morphologically, the neutrophils in CML appear normal. However, they are genetically abnormal and lack a normal neutrophil enzyme called **leukocyte alkaline phosphatase** (LAP). This is mostly tested to exclude secondary causes of neutrophilia (eg, infection) as well as the other myeloproliferative neoplasms, where the LAP is normal or increased.

CLINICAL CORRELATION

A decreased LAP score is not commonly used in routine clinical practice, given the use of the *BCR::ABL1* translocation for diagnosis.

What does a characteristic blood smear show in patients with CML?

Bone Marrow. The bone marrow in chronic-phase CML is moderately to markedly **hypercellular,** depending on how far the disease has progressed (Figure 3).

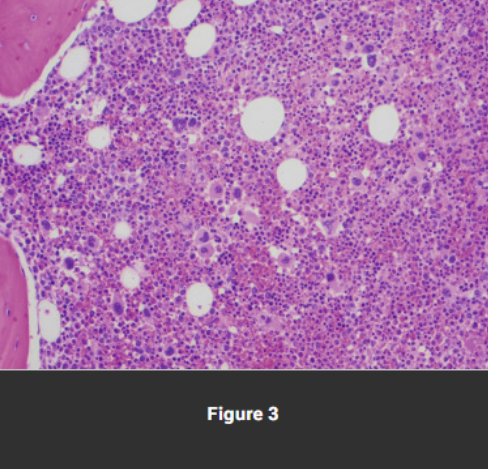

Figure 3

What makes up this hypercellularity? Neutrophils and their precursors predominate. Red blood cell precursors are present, but they are fewer in number than are the neutrophilic cells. You’ll have to kind of take our word for it, because until you have seen a lot of bone marrow biopsies, it’s hard to tell at this power what’s goi lls and not a lot of fat! There are also numerous megakaryocytes, many of which are characteristically small with hypolobated nuclei.

What is the consequence of this bone marrow hypercellularity? In chronic-phase CML, neoplastic cells squeeze out the few normal hematopoietic cells. As the disease progresses, neoplastic cells increase, whereas the RBC, normal leukocyte, and platelet counts drop. This happens slowly. By comparison, in acute leukemia, the counts can vary wildly: the WBC count may double in a few days, and the platelet count plummets.

Blast Crisis. This phase is basically the same thing as an acute leukemia: the WBC count rises rapidly, and the blast count is now more than 20%. Most of the time, the blast crisis is myeloid, but approximately 30% of the time, the blasts are lymphoblasts. This seems strange until we remember that in some cases of myeloproliferative disorders, the initial genetic mutation occurs in a stem cell that is so early and undifferentiated that it hasn’t even decided whether it wants to be lymphoid or myeloid. Therefore in blast crisis, lymphoblastic leukemia can occur. The blast phase is extremely difficult to treat, and survival is generally weeks to a few months.

Complications

The explosive WBC proliferation causes most of the CML complications. Potential problems include the following:

- Retrosternal pain from bone marrow expansion
- Splenic enlargement from extramedullary hematopoiesis
- Gouty arthritis caused by the high production of uric acid by fast turnover and lysis of the many WBCs in the circulation (the uric acid escapes from inside the cell in

Increased numbers of which type of cell in the blood or bone marrow largely determine that a patient with CML has progressed to blast crisis?

How Do We Diagnose Chronic Myeloid Leukemia (LO #3)?

The diagnosis of CML is first made by seeing the typical findings in the blood and bone marrow. That includes a CBC, peripheral smear, and bone marrow examination.

Complete Blood Count

An initial CBC will show elevated myeloid white blood cells, including neutrophils, eosinophils, and basophils (the specific increase will also depend on the subtype primarily affected).

CLINICAL REASONING

CML often presents with an increased white blood cell (WBC) count, with left-shifted maturation and increased basophils and eosinophils.

An important differential diagnosis with CML is a leukemoid reaction, which is an increase in neutrophils (usually with left-shifted maturation) that occurs because of an infection or a disease other than cancer.

How do we differentiate them? A leukemoid reaction can have increased WBC count (but usually <100,000), does not have prominent basophilia, and returns to normal following treatment of the infection/disease. Ultimately, if there is still a suspicion of CML, testing for t(9;22) is essential for the diagnosis.

Not all leukocytosis is from leukemia! Check the serum leukocyte alkaline phosphatase (LAP) levels if secondary leukocytosis is suspected, for example, from infection or another myeloproliferative neoplasm. In these secondary causes, the LAP level is increased, but the level will be low in CML.

Platelets are normal or increased, whereas erythrocytes will be normal. As the disease progresses, levels of thrombocytes will slowly decrease, and half of patients develop normocytic anemia.

Peripheral Smear

Peripheral smear is important in discerning the different WBC subtypes and cell maturity. For example, blast counts will increase during the blast crisis phase.

Bone Marrow Biopsy

Bone marrow examination can reveal percentage of blasts, bone marrow fibrosis, and mutated cell lines. Mutated cells can be screened for genetic aberrations.

CLINICAL CORRELATION

A diagnosis of CML can be made on peripheral blood. However, providers typically perform bone marrow biopsies as well to establish the phase of disease at diagnosis (chronic vs blast phase) and to monitor disease (monitoring percentage of blasts, presence or absence of marrow fibrosis).

Genetic Tests

The diagnosis is confirmed by the demonstration of the Philadelphia chromosome, the BCR::ABL1 fusion gene or the BCR::ABL1 fusion messenger RNA (mRNA). Testing can include cytogenetics, fluorescence in situ hybridization (FISH) analysis, or reverse transcription polymerase chain reaction (RT-PCR).

CLINICAL CORRELATION

In routine clinical practice, patients will usually receive all three tests (FISH, karyotype, and BCR

Doesn’t that seem redundant? What does each test tell you?

- FISH for t(9;22): This test is used because it’s fast but it only tells you about this translocation (whether it’s positive or negative); it does not look at any other chromosomes

- Karyotype: karyotype will also show the t(9;22) but, by visualizing all chromosomes, it can identify additional abnormalities that may create a worse prognosis. This test also takes longer than FISH (may be up to 1-2 weeks).

- PCR for BCR::ABL: This test usually takes a bit longer than FISH but establishes a baseline quantitative level of BCR::ABL that is monitored during the course of the patient’s treatment

How Do We Treat Chronic Myeloid Leukemia? (LO#4)

We use **tyrosine kinase inhibitors (TKIs)** to treat CML because they block the *BCR::ABL* fusion tyrosine kinase. This medication class was discussed previously in the “Targeted Inhibitors used in Cancer Therapy” brick. This targeted drug therapy has transformed CMLs from what used to be a 5-year survival into a chronic treatable condition, at least for those patients who respond to TKIs.

In practice, selection of a TKI is guided by disease phase and CML risk score. Other considerations include side effect profile, comorbid illnesses, medications that may interact with TKIs, availability, cost, and patient preference/convenience. However, most often, **imatinib** is the first choice for patients with CML. Imatinib binds to the kinase site of the BCR::ABL gene competitively inhibit phorylation of downstream signaling mediated by the activation of RTK is inhibited. Recall the mechanism of TKIs on the right side of Figure 4.

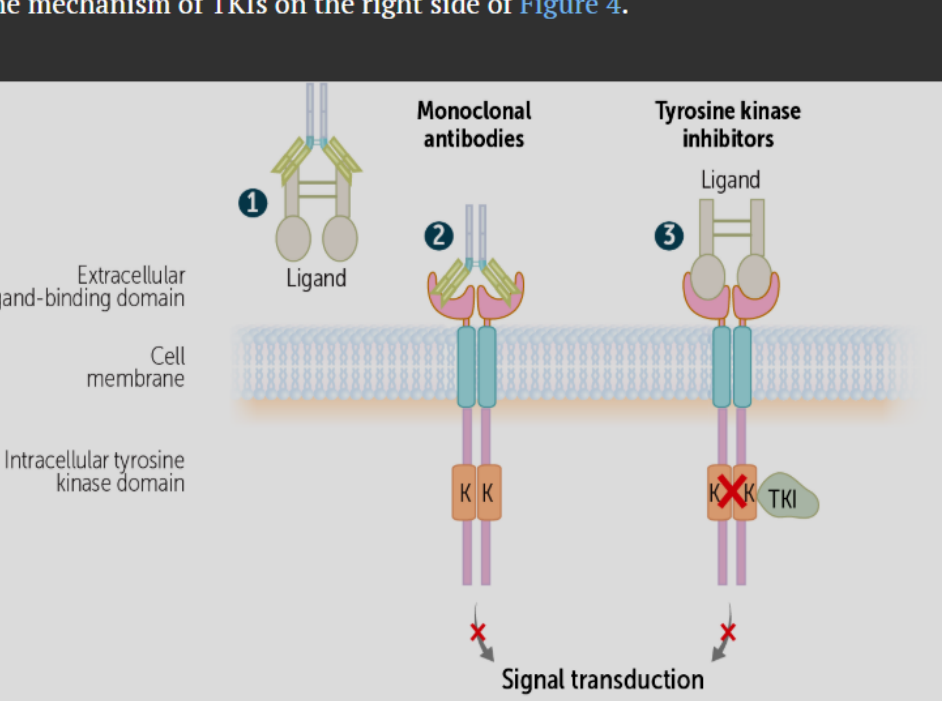
Monoclonal antibodies
Tyrosine kinase inhibitors
Ligand
1
2
3
Ligand
Extracellular ligand-binding domain
Cell membrane
Intracellular tyrosine kinase domain
K K
TKI
Signal transduction cascades blocked

Figure 4

All TKIs are associated with common, generally mild toxicities (eg, rash, nausea, edema, fatigue, myalgias/arthralgias) in the first months of treatment; most resolve spontaneously or can be controlled by dose adjustments. In addition to those listed above, imatinib is also associated with significant *GI side effects, such as nausea, vomiting, diarrhea, and abdominal pain/ distension.*

After initial treatmen d, other TKIs, such as **dasatinib** and **ponatinib**, may be tried. Why make a switch to a different TKI? Dasatinib and ponatinib, unlike imatinib, bind to both the active and inactive conformations of the *ABL* kinase domain and can *overcome imatinib resistance resulting from mutations in the BCR::ABL kinase*. Or, some patients may not tolerate therapy, due to the adverse effects, and the switch to another TKI is warranted. The general TKI adverse effects mentioned above apply to dasatinib and ponatinib, however, some specific effects include: pleural effusion and prolonged QTc (dasatinib) and thrombosis and liver toxicity (ponatinib).

In later stages of the disease, if TKIs have already been tried or are ineffective, less-specific chemotherapeutic drugs may be considered. This might include interferon alpha, hydroxyurea, or other chemotherapies. Younger patients may benefit from bone marrow or stem cell transplants.

CLINICAL CORRELATION

The TKIs mentioned in this brick are the ones most commonly used for CML, however, there are many other TKIs available. As mentioned above, selection of a TKI is guided by disease phase, CML risk score, AND side effect profile/ drug-interactions. Other TKIs that you may see include nilotinib, bosutinib, and asciminib.

Why are tyrosine kinase inhibitors effective in treating CML?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to LL, what clues would you expect to see on the CBC when you review LL’s record? How do you answer LL’s question?

You see that LL’s CBC showed a neutrophilic leukocytosis and basophilia. This, along with his bone marrow results and the presence of the Philadelphia chromosome, led to the diagnosis of CML, chronic phase. You tell LL that he understood your partner correctly. And when those parts of the two chromosomes fuse, a protein becomes synthesized (the “monster”) that leads to uncontrolled cell growth of white blood cells. The treatment you prescribe will inhibit the production of this protein, returning cell growth to normal. You tell LL that his disease can be controlled for years with this treatment.

Summary

- Chronic myeloid e neoplasm characterized by the overproduction of myeloid cells, particularly neutrophils.
- CML is typically diagnosed through blood and bone marrow tests, revealing high counts of neutrophils and variable percentage of blasts, and confirmed by genetic analysis showing the presence of the Philadelphia chromosome.
- All cases of CML are marked by the presence of the Philadelphia chromosome, generated via t(9;22). This joining of unrelated genes causes abnormal cell function, such as increased cell proliferation and defective DNA repair.
- CML is divided into chronic and blast phases. Chronic phase usually progresses slowly; patients may be asymptomatic or have symptoms of fatigue, early satiety, and abdominal pain (due to splenomegaly). Chronic phase is marked by <5% blasts. Blast phase, marked by >20% blasts, has similar symptoms to acute leukemia, with bone pain, night sweats, fever, weight loss, bleeding, and fatigue.
- Symptoms of CML can range from being asymptomatic to experiencing fatigue, splenomegaly, weakness, and infections.
- As CML progresses, patients may develop thrombocytopenia and anemia, leading to bleeding tendencies and fatigue respectively.
- Treatment for CML often involves the use of tyrosine kinase inhibitors, with stem cell or bone marrow transplant sometimes used in advanced disease.
- If left untreated, the life expectancy for CML patients is approximately 5 years, but with treatment, CML can become a chronic manageable condition.

Review Questic

1. **Which of the following peripheral blood or serum findings is most likely in a patient with chronic-phase chronic myeloid leukemia?**

   A. ● Basophilia
   B. ○ Blasts >20%
   C. ○ Elevated leukocyte alkaline phosphatase
   D. ○ Leukopenia
   E. ○ Thrombocytopenia

Hide Explanation

The correct answer is basophilia (A). Patients with chronic myeloid leukemia (CML) always have a basophilia; in fact, it is virtually the only condition in which basophilia is seen. Blasts >20% (B) are seen in acute myeloid leukemia. Patients with CML may develop blast crisis, in which the blast count is >20%, but this question asks about chronic-phase CML. An elevated leukocyte alkaline phosphatase (C) level is seen when there is a benign increase in WBC counts, as in infection. In CML, the neutrophils express decreased amounts of leukocyte alkaline phosphatase (LAP), so the LAP level is low. Leukopenia (D) is not seen in CML (in fact, the opposite is true; a leukocytosis is seen). Thrombocytopenia (E) is not typically seen in chronic-phase CML. It may be seen in the accelerated phase or blast crisis, however.

2. **The t(9;22) translocation in chronic myeloid leukemia leads to the production of which of the following aberrant proteins?**

   A. ○ JAK2 protein
   B. ○ p53 protein
   C. ○ RAS protein
   D. ○ Retinoblastoma protein
   E. ● Tyrosine kinase protein

Hide Explanation

The correct answer is a tyrosine kinase protein (E). In chronic myeloid leukemia (CML), the translocation between chromosome 9 and 22 causes a new fusion gene called *BCR::ABL* that encodes a constitutively active (“always on”) tyrosine kinase; this leads to uncontrolled cell proliferation. JAK2 protein (A) is mutated in several other myeloproliferative disorders, but not CML. p53 protein (B), RAS protein (C), and retinoblastoma, or retinoblastoma protein (D), are all mutated in a wide variety of cancers.

3. **Which adverse effect is most specifically associated with dasatinib?**

   A. ○ Liver toxicity
   B. ○ Periphera
   C. ● Pleural effusion
   D. ○ Severe GI bleeding

Hide Explanation

The correct answer is (C) Pleural effusion. Dasatinib is known for causing pleural effusions and can also prolong the QTc interval. (A) Liver toxicity is more specifically associated with ponatinib. (B) Peripheral neuropathy is not a hallmark side effect of TKIs; it’s more common with other chemotherapy classes. (D) GI side effects like nausea and diarrhea are common with imatinib, but not GI bleeding.

4. **A 60-year-old woman presents to her primary care physician for evaluation after routine laboratory studies reveal a white blood cell count of 120,000/µL. She reports mild fatigue but denies fever, weight loss, or night sweats. Physical examination reveals no lymphadenopathy but does show a palpable spleen tip below the left costal margin. Further testing reveals anemia and a platelet count of 450,000/µL. Peripheral blood smear shows numerous mature and immature granulocytes. Molecular testing reveals the presence of the BCR::ABL1 fusion gene. Which of the following best explains the natural course of this disease if left untreated?**

   A. ○ Rapid progression to marrow failure within weeks
   B. ● Stable dis transformation
   C. ○ Spontaneous remission in most patients
   D. ○ Immediate need for bone marrow transplantation
   E. ○ Development of pancytopenia due to marrow fibrosis

Hide Explanation

The correct answer is stable disease for several years followed by possible transformation (B). This patient has findings consistent with chronic phase chronic myeloid leukemia (CML), which is often diagnosed incidentally and may initially present with only mild symptoms like fatigue or splenomegaly. If untreated, the chronic phase typically lasts 3 to 4 years before potentially transforming into a blast phase, which resembles acute leukemia. Hence, the natural course involves relative stability followed by progression if not managed appropriately. Rapid progression to marrow failure within weeks (A) would be characteristic of blast phase of CML. Spontaneous remission in most patients (C) would not occur in CML; CML does not go away without treatment. Immediate need for bone marrow transplantation (D) is not required for the initial treatment of chronic phase CML. Typically, TKIs are started first as they have less side effects and often manage the disease appropriately. Patients with CML can develop fibrosis but not typically severe (E).

How would you rate this Brick?

References

- Gorczyca W. *Atlas of Differential Diagnosis in Neoplastic Hematopathology.* 4th ed. CRC Press; 2021.
- Greer JP, et al, eds. *Wintrobe’s Clinical Hematology.* 14th ed. Wolters Kluwer; 2018.
- https://onlinelibrary.wiley.com/doi/10.1002/ajh.26367
- https://www.uptodate.com/contents/clinical-features-and-diagnosis-of-chronic-lymphocytic-leukemia-small-lymphocytic-lymphoma?search=chronic%20lymphocytic%20leukemia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2
- https://www.uptodate.com/contents/overview-of-the-treatment-of-chronic-lymphocytic-leukemia?search=chronic%20lymphocytic%20leukemia&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- Liu D. *Tumors and Cancers: Endocrine Glands – Blood – Marrow – Lymph.* 1st ed. CRC Press; 2017.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differe yeloid leukemia (CML), delineating the chronic phase and the blast phase.~~

lymphoma?search=chronic%20lymphocytic%20leukemia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2
- https://www.uptodate.com/contents/overview-of-the-treatment-of-chronic-lymphocytic-leukemia?search=chronic%20lymphocytic%20leukemia&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- Liu D. *Tumors and Cancers: Endocrine Glands – Blood – Marrow – Lymph.* 1st ed. CRC Press; 2017.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of chronic myeloid leukemia (CML), delineating the chronic phase and the blast phase.~~

02 ☑ ~~Explain the pathophysiology of CML.~~

03 ☑ ~~Interpret clinical findings to properly diagnose CML.~~

04 ☑ ~~Outline the management plan of CML.~~

Objective Confidence: 4/4